Lab Plan – 6

Objective:

To develop a Flutter application that visually represents a xylophone with colorful bars and produces musical sounds when tapped.

For installations of dependencies visit: https://pub.dev/

Steps:

1. **Create a new Flutter project** named xylophone_app.

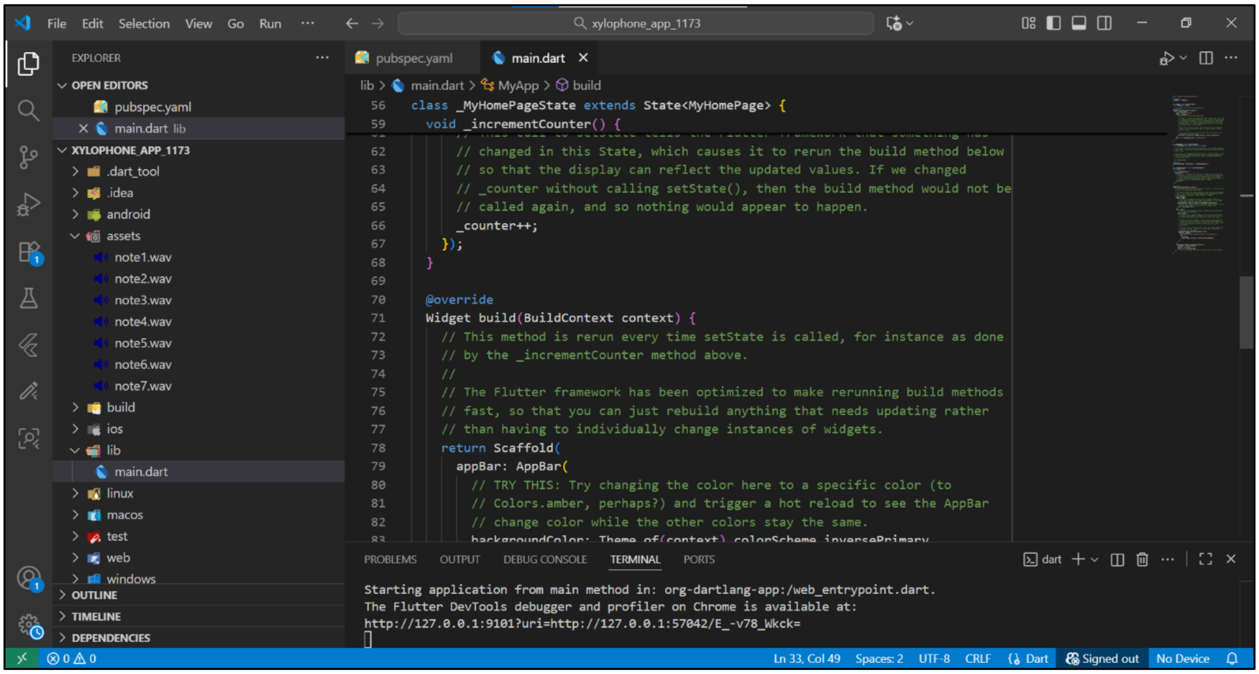
Code:

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab No 6',
      theme: ThemeData(
        // This is the theme of your application.
        //
        // TRY THIS: Try running your application with "flutter run". You'll see
        // the application has a purple toolbar. Then, without quitting the app,
        // try changing the seedColor in the colorScheme below to Colors.green
        // and then invoke "hot reload" (save your changes or press the "hot
        // reload" button in a Flutter-supported IDE, or press "r" if you used
        // the command line to start the app).
        //
        // Notice that the counter didn't reset back to zero; the application
        // state is not lost during the reload. To reset the state, use hot
        // restart instead.
        //
```

```dart
        // This works for code too, not just values: Most code changes can be
        // tested with just a hot reload.
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
      ),
      home: const MyHomePage(title: 'Xylophone App'),
    );
  }
}

class MyHomePage extends StatefulWidget {
  const MyHomePage({super.key, required this.title});

  // This widget is the home page of your application. It is stateful, meaning
  // that it has a State object (defined below) that contains fields that affect
  // how it looks.

  // This class is the configuration for the state. It holds the values (in this
  // case the title) provided by the parent (in this case the App widget) and
  // used by the build method of the State. Fields in a Widget subclass are
  // always marked "final".

  final String title;

  @override
  State<MyHomePage> createState() => _MyHomePageState();
}

class _MyHomePageState extends State<MyHomePage> {
  int _counter = 0;

  void _incrementCounter() {
    setState(() {
      // This call to setState tells the Flutter framework that something has
      // changed in this State, which causes it to rerun the build method below
      // so that the display can reflect the updated values. If we changed
      // _counter without calling setState(), then the build method would not be
      // called again, and so nothing would appear to happen.
      _counter++;
    });
  }

  @override
  Widget build(BuildContext context) {
    // This method is rerun every time setState is called, for instance as done
    // by the _incrementCounter method above.
```

```
    //
    // The Flutter framework has been optimized to make rerunning build methods
    // fast, so that you can just rebuild anything that needs updating rather
    // than having to individually change instances of widgets.
    return Scaffold(
      appBar: AppBar(
        // TRY THIS: Try changing the color here to a specific color (to
        // Colors.amber, perhaps?) and trigger a hot reload to see the AppBar
        // change color while the other colors stay the same.
        backgroundColor: Theme.of(context).colorScheme.inversePrimary,
        // Here we take the value from the MyHomePage object that was created by
        // the App.build method, and use it to set our appbar title.
        title: Text(widget.title),
      ),
      body: Center(
        // Center is a layout widget. It takes a single child and positions it
        // in the middle of the parent.
        child: Column(
          // Column is also a layout widget. It takes a list of children and
          // arranges them vertically. By default, it sizes itself to fit its
          // children horizontally, and tries to be as tall as its parent.
          //
          // Column has various properties to control how it sizes itself and
          // how it positions its children. Here we use mainAxisAlignment to
          // center the children vertically; the main axis here is the vertical
          // axis because Columns are vertical (the cross axis would be
          // horizontal).
          //
          // TRY THIS: Invoke "debug painting" (choose the "Toggle Debug Paint"
          // action in the IDE, or press "p" in the console), to see the
          // wireframe for each widget.
          mainAxisAlignment: MainAxisAlignment.center,
          children: <Widget>[
            const Text('You have pushed the button this many times:'),
            Text(
              '$_counter',
              style: Theme.of(context).textTheme.headlineMedium,
            ),
          ],
        ),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: _incrementCounter,
        tooltip: 'Increment',
        child: const Icon(Icons.add),
```

```
      ), // This trailing comma makes auto-formatting nicer for build methods.
    );
  }
}
```

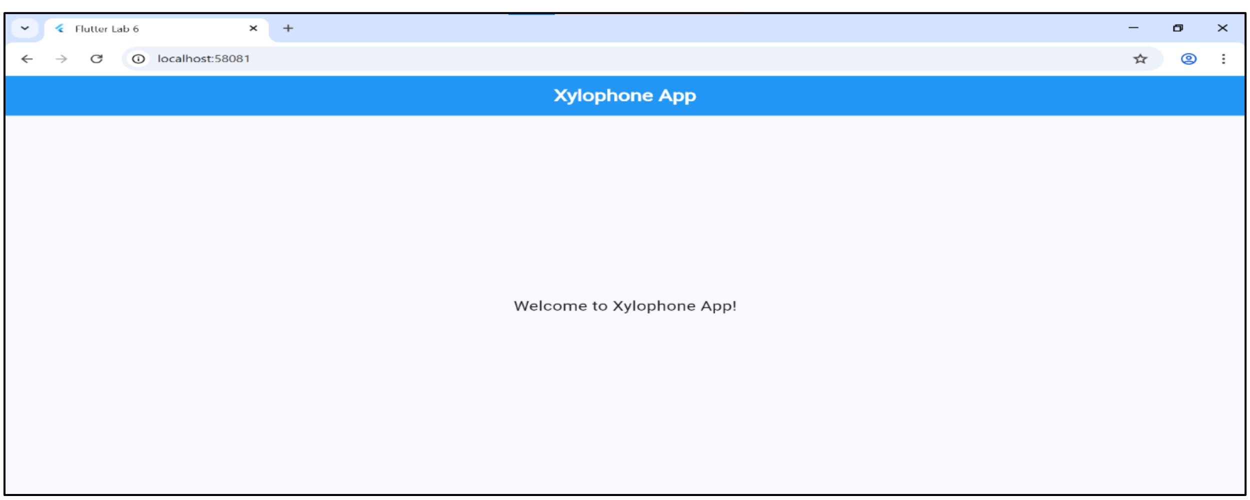


2. Create an **appbar** named as **"Xylophone"**

Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab 6',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),
      ),
      home: const MyHomePage(),
      debugShowCheckedModeBanner: false,
```

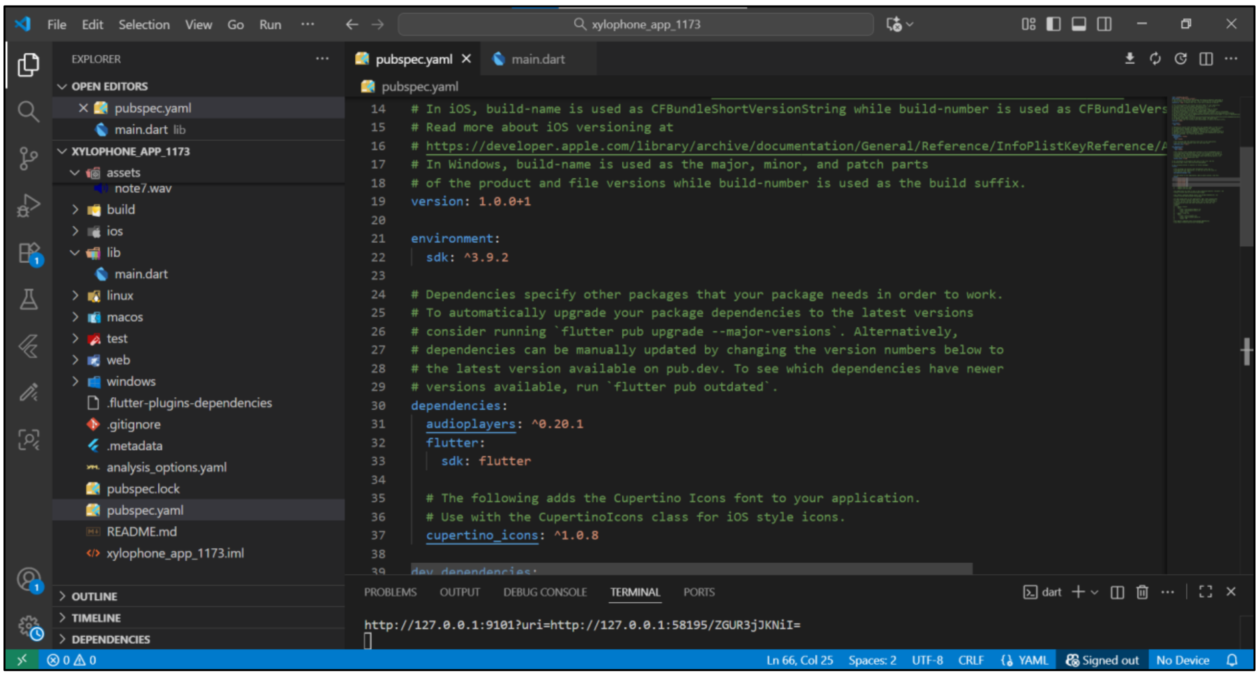
```
    );
  }
}

class MyHomePage extends StatelessWidget {
  const MyHomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        backgroundColor: Colors.blue,
        centerTitle: true,
        title: const Text(
          'Xylophone App',
          style: TextStyle(
            fontWeight: FontWeight.bold,
            fontSize: 22,
            color: Colors.white,
          ),
        ),
      ),
      body: const Center(
        child: Text(
          'Welcome to Xylophone App!',
          style: TextStyle(fontSize: 18),
        ),
      ),
    );
  }
}
```



3. **Add dependency** audioplayers: ^0.20.1 in the pubspec.yaml file.

Code:

```
dependencies:
  audioplayers: ^0.20.1
  flutter:
    sdk: flutter
```



4. **Add audio files** (note1.wav to note7.wav) in an assets folder and declare them under the flutter: section of pubspec.yaml.

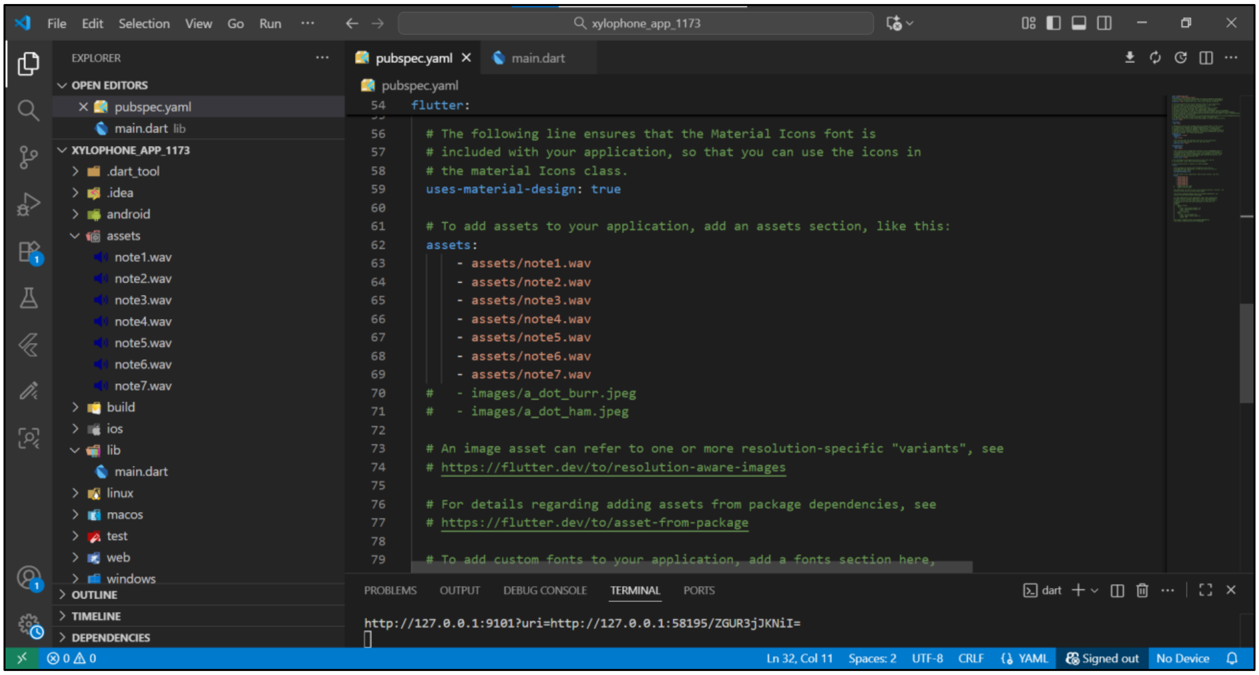
Code:

```
 # To add assets to your application, add an assets section, like this:
 assets:
   - assets/note1.wav
   - assets/note2.wav
   - assets/note3.wav
   - assets/note4.wav
```

- assets/note5.wav

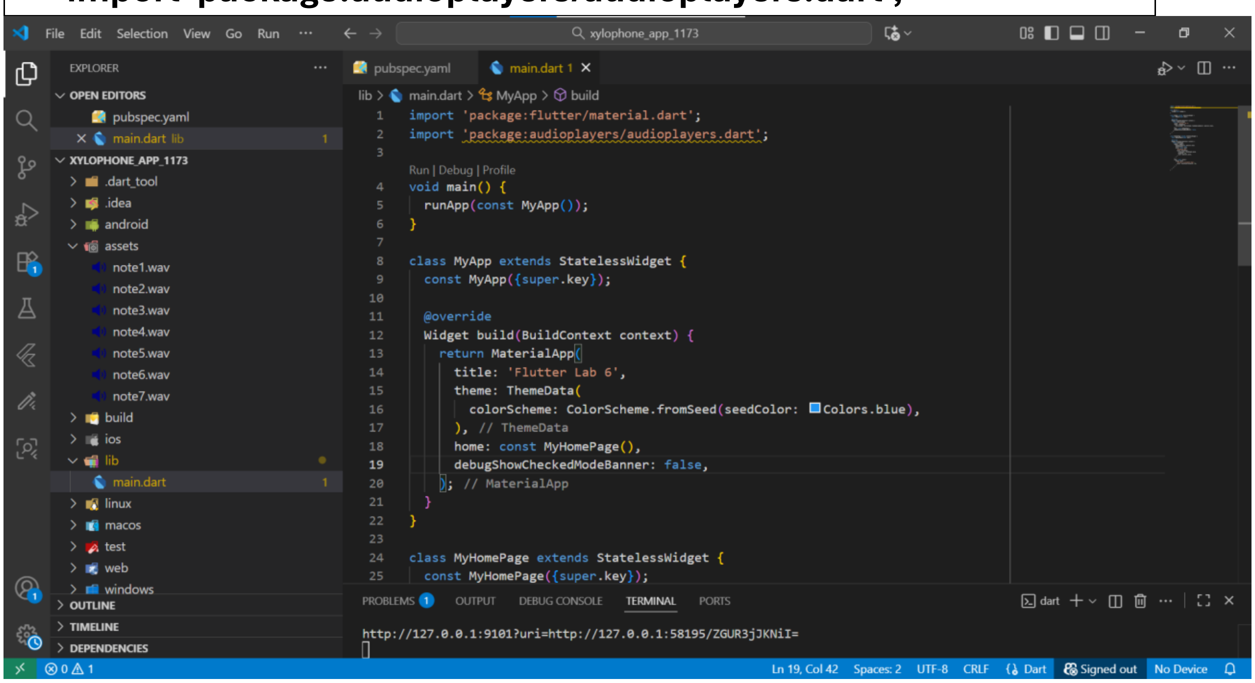
- assets/note6.wav

- assets/note7.wav



5. **Import package**:

import 'package:audioplayers/audioplayers.dart';

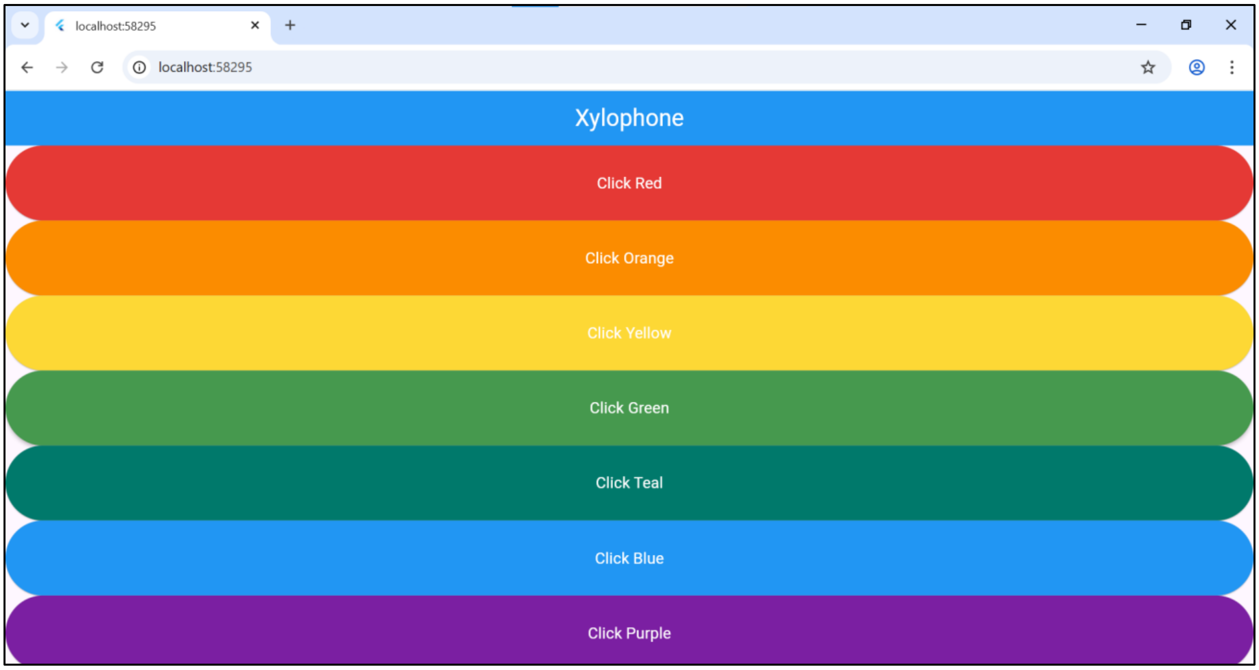


6. **Create 7 buttons** using TextButton widgets, each having:
    - A **unique color**
    - An **onPressed()** function that plays a corresponding sound.

Code:

```
import 'package:flutter/material.dart';
import 'package:audioplayers/audioplayers.dart';

void main() {
  runApp(const XylophoneApp());
}

class XylophoneApp extends StatelessWidget {
  const XylophoneApp({Key? key}) : super(key: key);

  // Helper to play a sound from assets/noteN.wav
  void playSound(int soundNumber) {
    final player = AudioCache();
    player.play('note$soundNumber.wav');
  }

  // Reusable key builder with rounded edges and no spacing
  Expanded buildKey(Color color, String label, int soundNumber) {
    return Expanded(
      child: Container(
        margin: EdgeInsets.zero, // no space between keys
        child: ElevatedButton(
          style: ElevatedButton.styleFrom(
            backgroundColor: color,
            shape: const StadiumBorder(), // rounded edges
          ),
          onPressed: () {
            playSound(soundNumber);
          },
          child: Text(
            label,
```

```dart
            style: const TextStyle(color: Colors.white, fontSize: 16),
          ),
        ),
      ),
    );
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: const Text(
            'Xylophone',
            style: TextStyle(fontSize: 24, color: Colors.white),
          ),
          centerTitle: true,
          backgroundColor: Colors.blue,
        ),
        body: SafeArea(
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.stretch,
            children: [
              buildKey(Colors.red.shade600, 'Click Red', 1),
              buildKey(Colors.orange.shade600, 'Click Orange', 2),
              buildKey(Colors.yellow.shade600, 'Click Yellow', 3),
              buildKey(Colors.green.shade600, 'Click Green', 4),
              buildKey(Colors.teal.shade700, 'Click Teal', 5),
              buildKey(Colors.blue.shade500, 'Click Blue', 6),
              buildKey(Colors.purple.shade700, 'Click Purple', 7),
            ],
          ),
        ),
      ),
    );
  }
}
```



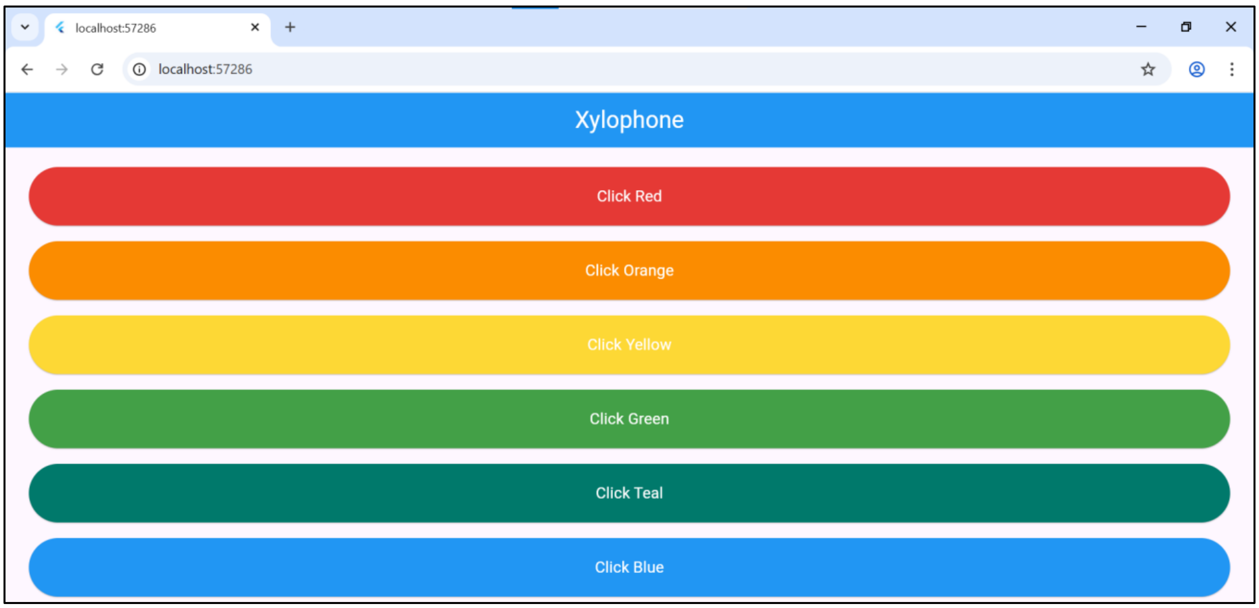
7. **Use Expanded widgets** so each button takes equal vertical space.

Code:

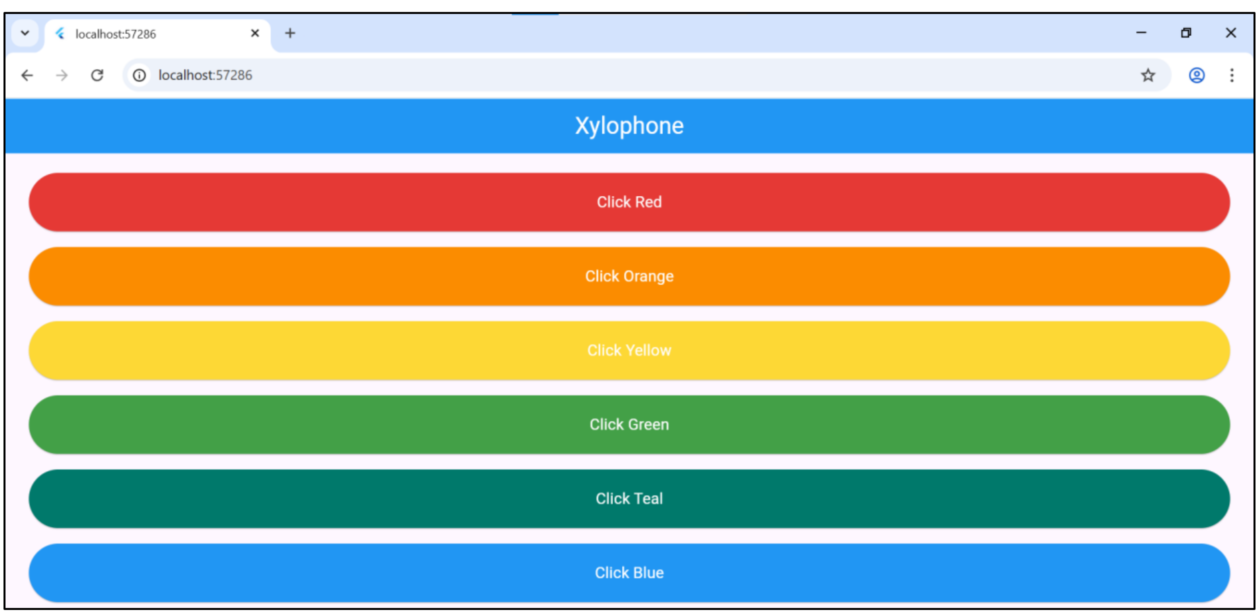
```
import 'package:flutter/material.dart';
import 'package:audioplayers/audioplayers.dart';

void main() {
  runApp(const XylophoneApp());
}

class XylophoneApp extends StatelessWidget {
  const XylophoneApp({Key? key}) : super(key: key);


  void playSound(int soundNumber) {
    final player = AudioCache();
    player.play('note$soundNumber.wav');
  }


  Widget buildKey(Color color, String label, int soundNumber) {
    return Padding(
      padding: const EdgeInsets.symmetric(vertical: 8.0),
      child: SizedBox(
        height: 60,
        width: double.infinity,
```

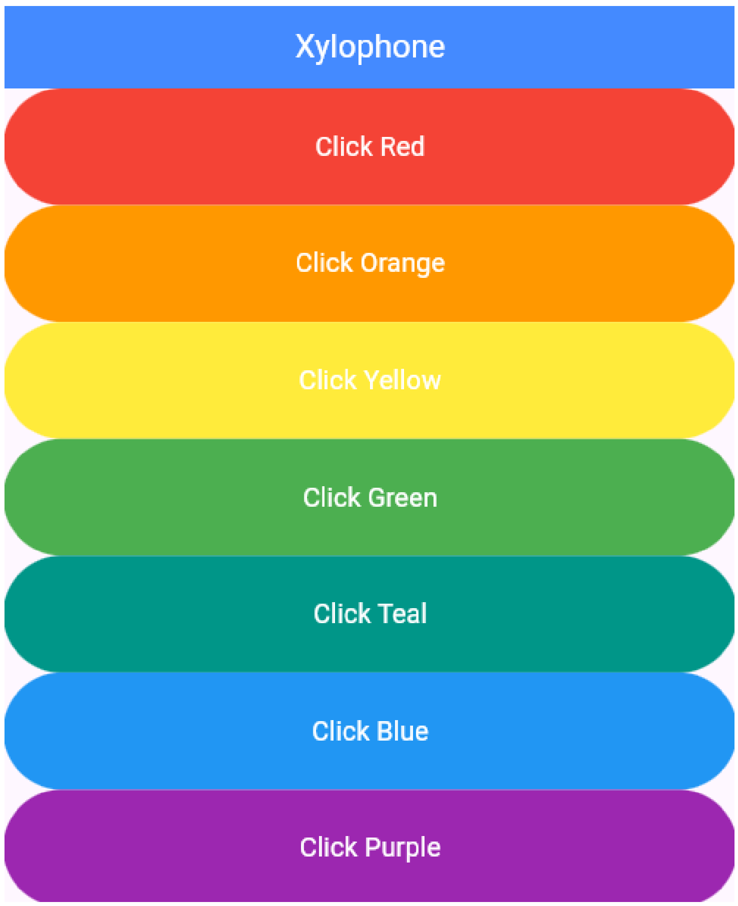
```
        child: ElevatedButton(
          style: ElevatedButton.styleFrom(
            backgroundColor: color,
            shape: const StadiumBorder(),
          ),
          onPressed: () {
            playSound(soundNumber);
          },
          child: Text(
            label,
            style: const TextStyle(color: Colors.white, fontSize: 16),
          ),
        ),
      ),
    );
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: const Text(
            'Xylophone',
            style: TextStyle(fontSize: 24, color: Colors.white),
          ),
          centerTitle: true,
          backgroundColor: Colors.blue,
        ),
        body: SafeArea(
          child: Padding(
            padding: const EdgeInsets.symmetric(
              horizontal: 24.0,
              vertical: 12.0,
            ),
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.stretch,
              children: [
                buildKey(Colors.red.shade600, 'Click Red', 1),
                buildKey(Colors.orange.shade600, 'Click Orange', 2),
                buildKey(Colors.yellow.shade600, 'Click Yellow', 3),
                buildKey(Colors.green.shade600, 'Click Green', 4),
                buildKey(Colors.teal.shade700, 'Click Teal', 5),
                buildKey(Colors.blue.shade500, 'Click Blue', 6),
```

```
                buildKey(Colors.purple.shade700, 'Click Purple', 7),
              ],
            ),
          ),
        ),
      ),
    );
  }
}
```



8. **Run the app** and test that each button plays its assigned sound.




Xylophone
Click Red
Click Orange
Click Yellow
Click Green
Click Teal
Click Blue
Click Purple